

Devon Whitaker

Research Scientist | Computational Biology

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PROFESSIONAL EXPERIENCE

Research Scientist, Institute for Biomedical Computing

Portland, OR | January 2024 – Present

Developing machine learning approaches for protein structure prediction validation within a team of 8 researchers. Implemented attention-based neural architectures to assess confidence scores on AlphaFold2 predictions across diverse protein families. Evaluated models on CASP14 and CAMEO benchmark datasets, achieving mean absolute error improvements of 12% relative to baseline geometric methods. Contributed to three peer-reviewed publications on uncertainty quantification in structural biology, with cumulative citation count of 47 across venues including ICML and Proteins journal.

PhD Candidate, Computational Molecular Biology Lab, University of Washington

Seattle, WA | August 2020 – December 2023

Doctoral research focused on generative models for molecular dynamics simulation. Designed graph neural network architectures to capture long-range atomic interactions and trained on trajectory data from the MD17 dataset. Published results in Nature Machine Intelligence (18 citations) and Bioinformatics (12 citations) demonstrating 2.3x faster inference compared to traditional force fields on validation benchmarks. Thesis defended with distinction on deep learning for protein dynamics prediction.

Research Assistant, Structural Genomics Center

Urbana, IL | June 2019 – July 2020

Supported computational analysis of cryo-EM structures and assisted with benchmark development for local atomic coordinate prediction. Implemented data preprocessing pipelines and contributed supplementary analyses to one first-author publication in Computational Biology Letters (8 citations).

EDUCATION

Ph.D. in Computational Biology, University of Washington, 2023

B.S. in Biophysics, University of Illinois, 2019